

Research Statement Outline

Probing the State: Mechanistic Interpretability for Secure State-Space Models

Ismail AISSAOUI | State Engineer in Artificial Intelligence

1. Introduction and Problem Statement

Large Language Models (LLMs) often exhibit a susceptibility to "learning" false information during training or fine-tuning. While current **Mechanistic Interpretability** research has made strides in understanding circuit-based logic in Transformers, the move towards high-performance architectures like **State Space Models (SSMs)** introduces new complexities. My research goal is to develop a mechanistic framework that probes how "factuality" vs. "falsehood" is represented in the compressed latent states of SSMs, specifically **Mamba**, to develop robust mitigation protocols.

2. Proposed Research Axes

2.1. 1. Latent State Attribution for Factual Consistency

In a Transformer, factual associations are often attributed to specific MLP layers or attention heads. In an SSM, information is stored in a continuously evolving hidden state. I propose to develop **Attribution Methods** for state-space transitions. By identifying which components of the state vector h_t are responsible for storing contradictory or false information, we can develop "surgical" intervention methods to prune or neutralize these specific dimensions without degrading the model's general performance.

2.2. 2. Mechanistic Comparison: Transformers vs. SSMs

Does the linear recurrence of an SSM make it more or less susceptible to "hallucinations" than the global attention of a Transformer? I intend to conduct a comparative mechanistic study. Using "Counterfactual Intervention" (changing a specific token and observing the state evolution), I will analyze how false information "leaks" into the model's long-term memory. This will lead to the design of safer training objectives that penalize specific types of state-space dynamics associated with misinformation.

2.3. 3. Security Envelopes for Reliable Inference

Inspired by my work on **Physics-Informed Neural Networks**, I propose the concept of **"Logical Validity Envelopes"**. By monitoring the state-space trajectories of an LLM during inference, we can identify "unphysical" or "illogical" transitions that correlate with the generation of false information. This real-time monitoring system would act as a security layer, flagging responses where the mechanistic behavior of the model deviates from known factual patterns.

3. Alignment with CopeNLU and DIKU

This research aligns with the lab's mission to build a mechanistic framework for LLM security. By bringing my expertise in **Advanced Sequential Modeling** and **Industrial Reliability**, I aim to contribute to a more theoretically grounded and secure generation of language models at the University of Copenhagen.